## Method of expressing concentration of solution

41. Two solutions of a substance (non electrolyte) are mixed in the following manner 480 *ml* of 1.5*M* first solution + 520 *mL* of 1.2*M* second solution. What is the molarity of the final mixture
(a) 1.20 *M* (b) 1.50 *M*
(c) 1.344 *M* (d) 2.70 *M*

42. The normal amount of glucose in $100ml$ of blood (8–12 hours after a meal) is
(a) $8mg$ (b) $80mg$
(c) $200mg$ (d) $800mg$

43. Molar solution means 1 mole of solute present in
(a) 1000*g* of solvent
b) 1 litre of solvent
(c) 1 litre of solution
d) 1000*g* of solution

44. What will be the molality of a solution having $18g$ of glucose (mol. wt. = 180) dissolved in $500g$ of water
(a) $1m$ (b) $0.5m$
(c) $0.2m$ (d) $2m$

45. A solution of $Al_2(SO_4)_3\{d = 1.253gm/ml\}$ contain 22% salt by weight. The molarity, normality and molality of the solution is
(a) 0.805 *M*, 4.83 *N*, 0.825 *M*
(b) 0.825 *M*, 48.3 N, 0.805 *M*
(c) 4.83 *M*, 4.83 *N*, 4.83 *M*
(d) None

46. Which of the following should be done in order to prepare $0.40MNaCl$ starting with $100ml$ of $0.30MNaCl$ (mol.wt. of $NaCl = 58.5$)
(a) Add $0.585gNaCl$
b) Add $20ml$ water
(c) Add $0.010mlNaCl$
d) Evaporate $10ml$ water

47. Which of the following solutions has the highest normality
(a) 8 *gm* of $KOH$ / litre
b) $N$ phosphoric acid
(c) 6 *gm* of $NaOH$ / 100 *ml*
d) $0.5MH_2SO_4$

48. What volume of $0.8M$ solution contains 0.1 mole of the solute
(a) $100ml$ (b) $125ml$
(c) $500ml$ (d) $62.5ml$

49. Hydrochloric acid solution $A$ and $B$ have concentration of $0.5N$and $0.1N$ respectively. The volumes of

solutions $A$ and $B$ required to make $2 litres$ of $0.2N\ HCl$ are

(a) $0.5l$ of $A + 1.5l$ of $B$

(b) $1.5l$ of $A + 0.5l$ of $B$

(c) $1.0l$ of $A + 1.0l$ of $B$

(d) $0.75l$ of $A + 1.25l$ of $B$

**50.** Conc. $H_2SO_4$ has a density of 1.98 *gm/ml* and is 98% $H_2SO_4$ by weight. Its normality is

(a) 2 *N* (b) 19.8 *N*

(c) 39.6 *N* (d) 98 *N*

**51.** The mole fraction of the solute in one molal aqueous solution is

(a) 0.027 (b) 0.036

(c) 0.018 (d) 0.009

**52.** With 63 *gm* of oxalic acid how many *litres* of $\frac{N}{10}$ solution can be prepared

(a) 100 *litre* (b) 10 *litre*

(c) 1 *litre* (d) 1000 *litre*

**53.** Molarity of $0.2N H_2SO_4$ is

(a) 0.2 (b) 0.4

(c) 0.6 (d) 0.1

**54.** 10.6 grams of a substance of molecular weight 106 was dissolved in $100ml$. $10ml$ of this solution was pipetted out into a $1000ml$ flask and made up to the mark with distilled water. The molarity of the resulting solution is

(a) $1.0 \rightleftarrows M$ (b) $10^{-2}M$

(c) $10^{-3}M$ (d) $10^{-4}M$

**55.** The mole fraction of water in 20% aqueous solution of $H_2O_2$ is

(a) $\frac{77}{68}$ (b) $\frac{68}{77}$

(c) $\frac{20}{80}$ (d) $\frac{80}{20}$

**56.** Mole fraction $(X)$ of any solution is equal to

(a) $\frac{\text{No. of moles of solute}}{\text{Volume of solution in litre}}$

(b) $\frac{\text{No. of gram equivalent of solute}}{\text{Volume of solution in litre}}$

(c) $\frac{\text{No. of moles of solute}}{\text{Mass of solvent in kg}}$

(d) $\frac{\text{No. of moles of any constituent}}{\text{Total no. of moles of all constituents}}$

**57.** When $W_B gm$ solute (molecular mass $M_B$ ) dissolves in $W_A gm$ solvent. The molality $M$ of the solution is

(a) $\frac{W_B}{W_A} \times \frac{M_B}{1000}$ (b) $\frac{W_B}{M_B} \times \frac{1000}{W_A}$

(c) $\frac{W_A}{W_B} \times \frac{1000}{M_B}$ (d) $\frac{W_A \times M_B}{W_B \times 1000}$

**58.** Normality $(N)$ of a solution is equal to

(a) $\frac{\text{No. of moles of solute}}{\text{Volume of solution in litre}}$

(b) $\frac{\text{No. of gram equivalent of solute}}{\text{Volume of solution in litre}}$

(c) $\frac{\text{No. of moles of solute}}{\text{Mass of solvent in kg}}$

(d) None of these

**59.** The volume strength of $1.5NH_2O_2$ solution is

(a) 4.8 (b) 5.2

(c) 8.8 (d) 8.4

**60.** How many $gm$ of $H_2SO_4$ is present in $0.25gm$ mole of $H_2SO_4$

(a) 24.5 (b) 2.45

(c) 0.25 (d) 0.245

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